

NAFIZ IMTIAZ

Queens, New York

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PROFILE SUMMARY

Operations research analyst and AI professional focused on trustworthy decision systems for U.S. education, workforce development, and organizational planning. Work combines predictive analytics, SQL-based integration of fragmented data, model validation, responsible AI, small language models, and resource-aware decision support. Research spans AI-enabled education, governance-ready analytics, federated artificial intelligence of things (AIoT), and public-sector interoperability, including NSF-supported work; M.S. in Business Analytics (3.91 GPA) and IEEE Best Paper Award recipient.

CORE COMPETENCIES

Predictive Analytics | SQL and Data Integration | Dashboards and Reporting | AI Product Requirements | UAT and Model Validation | Learning and Training Analytics | Responsible AI and Governance | Process Mining

EDUCATION

M.S. in Business Analytics (GPA: 3.91), Montclair State University

Montclair, NJ

B.B.A. in Management Information System (MIS), North South University

Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

Operations Research Analyst

July 2026 - Present

UpSkill Consultancy Inc.

Jackson Heights, NY

- Conduct operations research studies of education- and workforce-training business processes, developing mathematical, statistical, and optimization models that support operational planning and organizational performance.
- Collect, validate, and analyze large, fragmented datasets from multiple sources to identify trends, patterns, and improvement opportunities, documenting analytical methodologies, assumptions, and models for transparent human review.
- Prepare analytical reports, dashboards, forecasts, and visualizations that translate research findings into data-driven recommendations for operational efficiency and continuous improvement.

Business Analyst - AI Products and Data Analytics

February 2026 – July 2026

Bits and Binaries, Inc.

Irving, TX

- Contribute to local LLM and AI-enabled analytics products for forecasting, retention analysis, and decision support; translate stakeholder needs into analytics requirements, user stories, reporting specifications, and validation criteria.
- Develop, optimize, and validate SQL-based data pipelines and dashboards for incomplete multi-source datasets, including internal upskilling analytics that monitor learning progress, skill gaps, and training-support needs.

Business Analyst - Product Research and Development

October 2025 - February 2026

DCITM LLC

Richmond, VA

- Supported AI automation and SQL-based data integration involving training activity, assessment progress, support records, and business-performance indicators for workforce-learning analytics.
- Gathered requirements, evaluated workflows, prepared documentation and visualizations, and synthesized research on agentic AI architectures for product development and automated decision-support pipelines.

Consultant Business Analyst

February 2025 - October 2025

NetCom Learning Global

Manhattan, NY

- Contributed to N+, an AI-powered adaptive learning application, through product design, data requirements, analytics integration, and cross-functional coordination.
- Developed visual analyses of operations, sales, and stakeholder requirements and consulted with professionals to shape customized learning solutions.

Project Research Assistant

January 2019 - March 2022

BacBon Limited

Dhaka, Bangladesh

- Supported BacBon School and BacBon Tutors by collaborating with software teams under agile workflows to improve education-platform performance and user experience; also delivered remote English instruction through the SLC platform in collaboration with Shinkoshuppansha Keirinkan Co., Ltd.
- Collaborated on Asian Development Bank-supported Math and English study materials for junior-high learners and supported international education content and program coordination.

SELECTED TECHNICAL PROJECT AND CERTIFICATION

Process Mining with Celonis - Montclair State University

December 2024

- Analyzed and redesigned a complex business process using Celonis, identified structural inefficiencies, and developed recommendations for concurrent and more efficient workflows.
- Celonis Academy certification: Academic Process Mining Fundamentals.

FedCuraLM - Montclair State University

March 2026

- Coauthored the research paper FedCuraLM: Reliability-Calibrated and Resource-Aware Label Auditing with Small Language Models for Non-IID Federated AIoT, developed in collaboration with Illinois Wesleyan University, Illinois State University, and Heartland Community College under NSF S-STEM Award No. 1742224.

- Contributed to the design of the framework's reliability-calibration and evaluation methodology, including structured Semantic Evidence Cards, model abstention when evidence is insufficient, and calibration assessment using negative log-likelihood, Brier score, and expected calibration error.
- Helped develop the framework's auditable decision-support and data-management components, including an append-only ledger of evidence, review outcomes, and decision history, and resource-aware allocation of small language models according to case difficulty.

EduWork DataBridge (Open Source) - GitHub

July 2026

- Designed and developed EduWork DataBridge, an open-source reference framework released under the Apache License 2.0 for governed learning and workforce data interoperability, including a public programming interface, synthetic demonstration datasets, and complete documentation, tests, and release verification for reproducible evaluation.
- Implemented safe ingestion and validation of fragmented records from human-resources, learning-management, assessment, and credential systems, transparent identity resolution with human-review boundaries, and documented, masked, lineage-aware exports for approved uses.

HONORS AND LEADERSHIP

Best Paper Award - IEEE CS BDC Symposium 2024

Dhaka, Bangladesh | November 2024

- Recognized for "Optimizing Stroke Prediction Models for Real-World Application: Tackling Data Imbalance Without Synthetic Samples."

LAMP (Learn, Act & Make Program) - Young Leader

Tokyo, Japan | February 2019

- Represented Bangladesh in a cross-cultural social-impact program focused on entrepreneurship, customer satisfaction, volunteer activities, and problem solving with participants and policy experts from different backgrounds.

PUBLICATIONS AND RESEARCH OUTPUTS

Peer-Reviewed Journal Articles

1. Imtiaz, N., Zannat, F., Tohfa, R., I., Mahmud, S., et. al. (2026). *A Simulation-Based Adaptive Learning Analytics Framework for Workforce Upskilling in U.S. Community College*. *International Journal of Modern Education and Computer Science* (Q3) (Accepted).
2. Imtiaz, N., Kundu, T. R., Roy, A., Bhuiyan, M. I. H., Rahman, K., & Islam, M. K. (2025). *Governance readiness beyond predictive performance: An empirical benchmark for higher-education early warning systems*. *Frontiers in Computer Science and Artificial Intelligence*, 4(5), 49-65. <https://doi.org/10.32996/fcsai.2025.4.5.6X>
3. Alim, I., Imtiaz, N., Prince, A. A., & Hasan, M. A. (2025). *AI and Blockchain Integration: Driving Strategic Business Advancements in the Intelligent Era*. (2025). *Journal of Engineering and Computational Intelligence Review*, 3(2), 38-50. <https://doi.org/10.63544/j4wz9h74>
4. Imtiaz, N., Zannat, F., Ahmed, S., Hasan, M. A., & Mahmud, S. (2025). *Leveraging AI for data-driven decision making and automation in the USA education sector*. *Journal of Economics, Management & Business Administration*, 4(1), 87-106. <https://doi.org/10.59075/jemba.v4i1.551>
5. Imtiaz, N., & Karim, M. Z. (2023). *Embracing data-driven process optimization to highly transform instructional administration in higher education institutions*. *Business and Social Sciences*, 1(1), 1-8. <https://doi.org/10.25163/business.1110399>
6. Choudhury, P., & Imtiaz, N. (2020). *Overcoming data excess to improve decision-making and information systems plans for organizational performance*. *Journal of Primeasia*, 1(3), 1-7. <https://doi.org/10.25163/primeasia.1110403>

Conference Papers & Abstracts

7. Hasan, M. A., Imtiaz, N., Ihsan, M. A., & Mahmud, S. (2026). *A multi-agent knowledge management framework for real-time U.S. Medicaid program integrity analytics: Scaling semantic interoperability*. Proceedings of the 2026 3rd IEEE International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE). <https://doi.org/10.1109/RMKMATE69073.2026.11518763> (Q1)
8. Ihsan, M. A., Eram, A. F., Imtiaz, N., Mahmud, A. S., Khadem, M. M., & Nahar, L. (2024). *Optimizing stroke prediction models for real-world application: Tackling data imbalance without synthetic samples*. IEEE CS BDC Symposium 2024, Vol. 3, November 22-23, 2024. Best Paper Award.

Submitted Papers

9. Bhuiyan, M. I. H., Hossain, M., I., Imtiaz, N., Zannat, F., Tohfa, R., I., & Islam, M., K. (2026). *From Risk Registers to Uncertainty Observatories: Managing Shadow Generative AI Innovation Under High Risk and High Uncertainty*. *Technology in Society*, Elsevier (Q1), (Submitted).
10. Rashid, M., T., Imtiaz, N., Bhuiyan, M. I. H., Zannat, F., Tohfa, R., I., Mahmud, S., et. al. (2026). *From prediction to support: An equity-centered adaptive learning analytics framework for workforce readiness* TechTrends, Springer (Q1), (Submitted).
11. Rashid, M., T., Ahmed, M., S., & Imtiaz, N. (2026). *FedCuraLM: Reliability-Calibrated and Resource-Aware Label Auditing with Small Language Models for Non-IID Federated AIoT*. *Computers*, mdpi (Q1), (Submitted).